

TOPIC 38/60

What is Pruning in LLMs?

Cutting the Deadwood.

How we slice away redundant parameters to speed up LLMs without losing their intelligence.

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THE PROBLEM

The Over-inflated Network

Deep Learning models are notoriously overparameterized. They contain billions of weights, but during inference, many of these weights remain stagnant or hold values near zero.

Computational Waste

Even if a weight is practically useless, the GPU is still forced to load it from memory and perform floating-point multiplications on it during every single token generation step.

Result: Major bottlenecks in memory bandwidth and inference speed.

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THE CONCEPT

Neural Gardening



What is Pruning?

Like pruning dead branches off a tree to make it healthier, **Pruning** in neural networks involves systematically removing weights or structural blocks that do not contribute significantly to the model's accuracy.

Original Model

Heavily redundant, highly demanding on hardware, slow memory retrieval.

Pruned Model

Streamlined, faster throughput, lower VRAM requirements, identical performance.

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THE METHODOLOGY

Two Ways to Trim

How we cut defines whether our hardware actually gets a speed boost:

1. Unstructured Pruning

Snips arbitrary individual weights anywhere in the matrix.

Con: Creates sparse matrices that standard GPUs struggle to speed up without special libraries.

2. Structured Pruning

Removes entire attention heads, channels, or layers wholesale.

Pro: Directly matches standard dense GPU calculations, giving an immediate hardware speed boost.

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THE RULES

Who Gets the Chop?

To find out which components are least important, we run criteria sweeps across the weights:

1. Magnitude-Based Pruning:

We look at the absolute value of each weight. Weights close to zero ($|w| \approx 0$) are considered dead weight and sliced first.

2. Activation-Aware (Wanda):

We evaluate weights based on the input activations. If a large weight is multiplied by an input that is consistently zero, it is pruned.

Modern approaches focus on keeping the remaining signals clean without destroying core circuits.

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MODERN LLM INNOVATIONS

One-Shot Solvers

Traditionally, pruning required retraining the model for days to recover accuracy. With 70B+ parameter models, retraining is far too expensive. This sparked one-shot pruning breakthroughs:

SparseGPT

Solves a local reconstruction problem to adjust the surviving weights to compensate for the pruned ones. Can achieve up to 50% sparsity in one shot.

Wanda

Prunes weights by multiplying their magnitude with input activations. Extremely fast, requires zero gradient optimization or retraining.

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THE 2:4 SPARSITY PATTERN



NVIDIA Ampere and Hopper GPUs feature dedicated hardware pipelines for semi-structured sparsity. They require exactly **two out of every four values** in a matrix to be zero.

How 2:4 Sparsity functions:

Step 1: The algorithm prunes the matrix until every block of 4 sequential weights has exactly 2 zeros.

Step 2: Tensor Cores ignore the zeros and pack the remaining 2 values into half the memory space.

Step 3: Math pipeline executes twice as fast, giving a direct 2x inference speedup.



EXECUTION MAP

The Pruning Workflow

How an LLM is safely pruned and optimized for production:

1. Analyze Profile

Pass calibration data to measure activation norms
2. Mask and Cut

Nullify weights matching the chosen pattern (e.g. 2:4)
3. Re-Weight & Fine-tune

Tweak surviving values to restore lost perplexity

A carefully planned workflow ensures that conversational nuance and reasoning capabilities aren't chopped away.



THE TOOLKIT

Model Compression Suite



Pruning

Physically discards useless weights/connections entirely. Changes the architecture design.



Quantization

Retains all connections, but lowers the numeric precision (e.g., from Float16 to 4-bit Integers).



Distillation

Transfers broad reasoning logic from a massive "Teacher" LLM to a distinct, smaller "Student" LLM.

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Model Pruning

CONCEPT UNLOCKED



Slices away inactive & redundant parameters



Uses structured groups to preserve GPU-friendliness



Leverages 2:4 sparsity for direct hardware speedups



One-shot solvers (Wanda/SparseGPT) require zero retraining

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